



CEBELÜ

RoboCupJunior Rescue Maze

We are a rookie team, we dont have any awards



The Cebelu soldiers were cavalry troops in the Ottoman Empire. They served under the timar system and joined the army during wars. Cebelu soldiers were equipped with horses and weapons provided by their timar holders. They played an important role in the Ottoman military for many years.

Hardware

ToF Sensors (8x) (Purple): Four side-mounted, two front-facing, and two diagonal ToF sensors are used for wall following, maze navigation, robot alignment, ramp detection, and obstacle detection. The diagonal sensors were positioned using trigonometric calculations to improve obstacle recognition and validate ramp identification.

IMU: Provides orientation and motion data, enabling more reliable alignment, precise turns, and stable movement on ramps.

Victim Cameras (Left & Right) (Black): Used to detect victims on both sides of the maze and determine the correct side for rescue kit delivery.

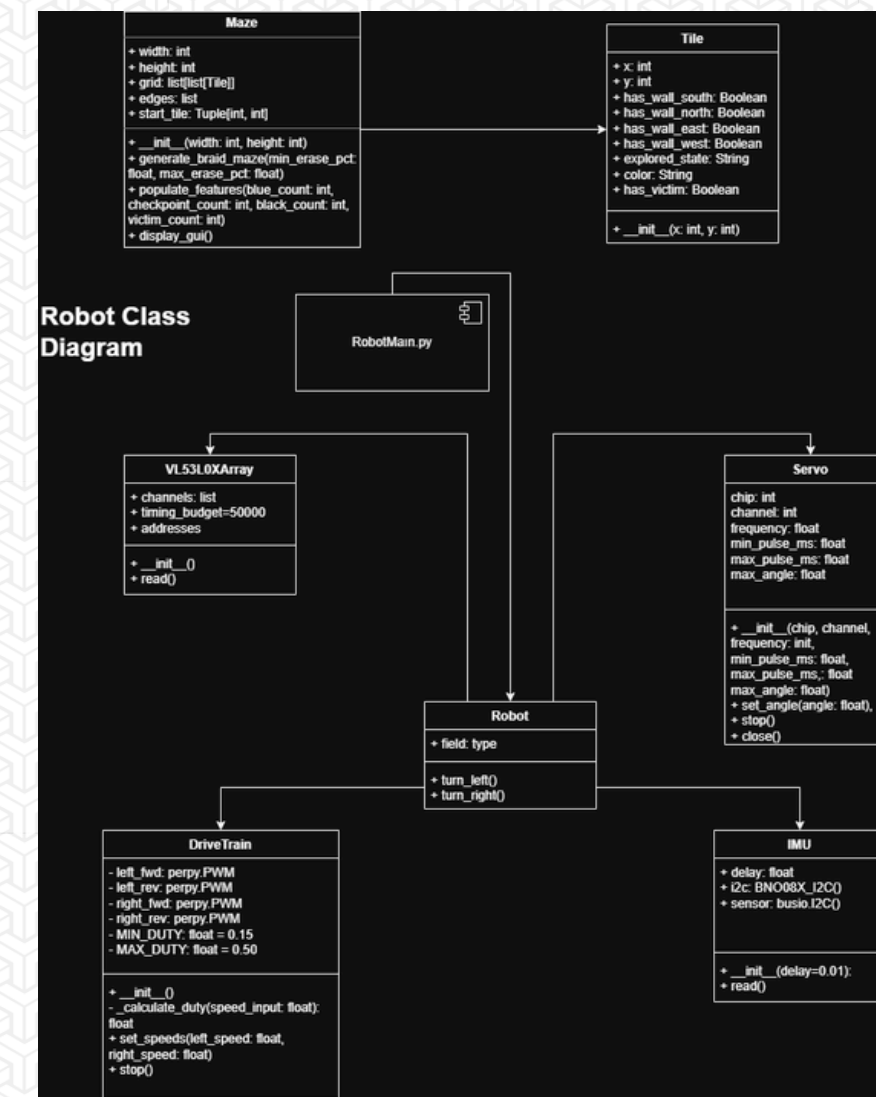
Color Sensor: Used to detect floor colors such as black tiles and other field markings.

Encoders: Measure wheel rotation for accurate distance tracking and movement control.

DC Gear Motors: Provide the driving force for the robot. Their speed and direction are controlled precisely to ensure smooth navigation, accurate turns, and reliable movement through the maze.



Software



Muhammed Hakan Özkan: Team Captain
Leads the team. Responsible for project planning, coordinating members and ensuring smooth communication. Also takes part in manufacturing and programming process. He loves to plan things.



Ömer Kaan Özden: 3D Designer and Mechanical Engineer
Responsible for designing and printing mechanical parts and mechanisms of robot. Takes part in manufacturing the robot. Also has unique singing and dancing skills we want him to perform.



Fevzi Gümüştekin: Electronics Engineer
Designs and builds electronic systems of the robot. Integrating software into system with Software Developer. He loves jumper wires



Mustafa Altan Dağ: Software Developer
Develops and programs robots algorithm and code. Integrating hardware into system with Electronics Engineer. He loves talking with people, we are sure he is going to socialize in the field.



Fatih Kürşat Cansu: Team Mentor
Provides guidance to the team. The adult individual responsible for ordering and tracking materials, required documents for regulation (deed of consent etc.), travel planning, and coordination of students while abroad.